CROPSAGE | TECHNICAL VALIDATION

FortyGuard Plainview Heatmap Test Report

All-crop multi-window validation: 7, 14, 21 and 30 days

4	88	64	0
TEST WINDOWS	CROP OUTCOMES	API RESPONSES	NULL VALUES

This report records the completed FortyGuard Temperature API validation for a Plainview, Texas demonstration farm. It covers the initial cotton test and Tests A-D across the full 22-crop catalog, with threshold exposure, persistence, activity traceability, interpretation and limitations.

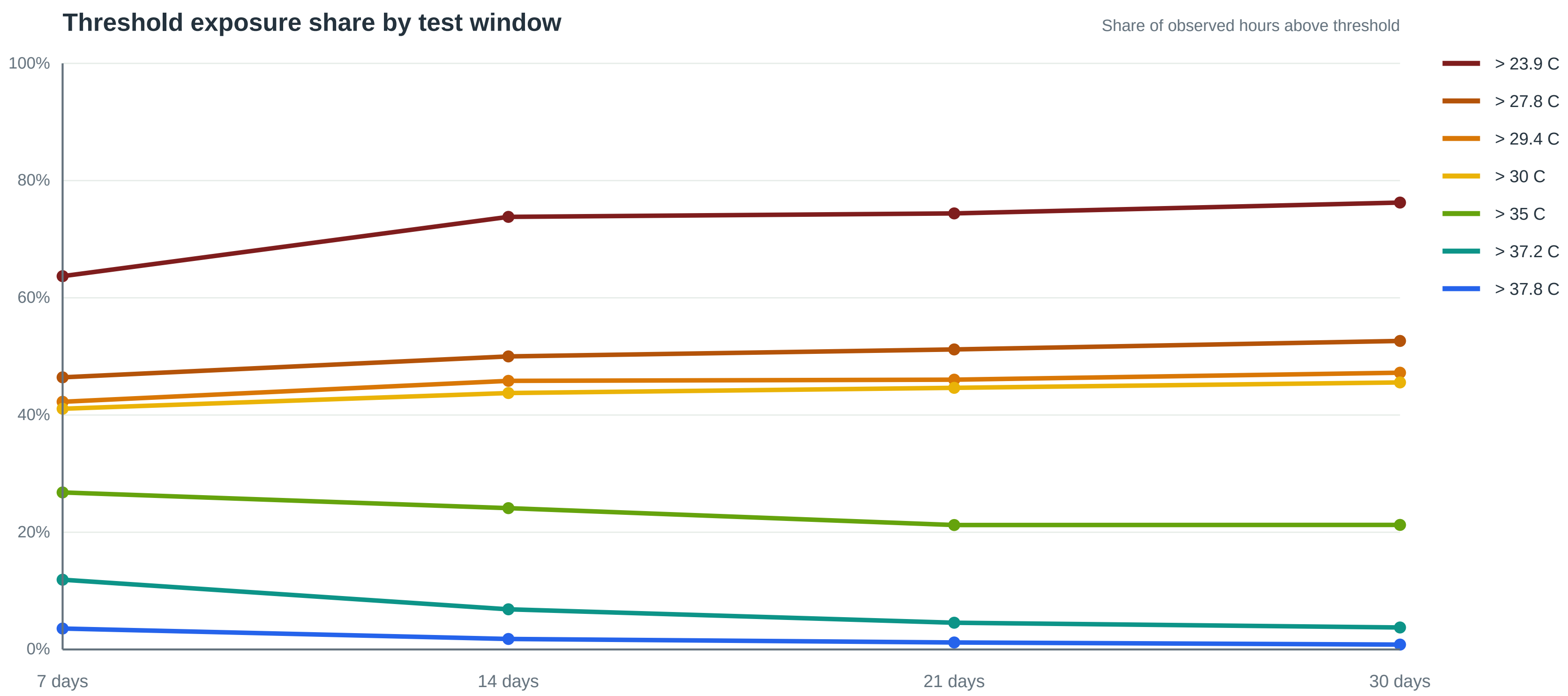
Prepared for the FortyGuard Global AI Hackathon '26

Report date: 27 August 2026 | Study area: Plainview, Texas

Executive Summary

64 / 64	4,352	88
VALIDATED API RESPONSES	NUMERIC CELL VALUES	CROP-WINDOW OUTCOMES

- All 64 FortyGuard responses completed successfully and returned 68 spatial cells each.
- The 22-crop catalog produced 88 crop-window outcomes across Tests A-D; 10 crops had numeric heat thresholds represented by seven unique threshold groups.
- Threshold results were spatially uniform across the 68 cells, so the evidence supports farm-level temporal heat screening, not within-field hotspot claims.
- Raw peak hour 15 is retained for traceability but excluded from scoring until the API timezone basis is confirmed.



Interpretation: lower heat thresholds naturally accumulate more exposure time. The hottest thresholds become proportionally rarer as the observation window expands, while the recurring 39.2 C maximum confirms that acute heat risk remains present.

Report status

- Test completed: 27 August 2026
- Test window: 19-25 August 2026
- Notebook: `fortyguard_heatmap_test.ipynb`
- Saved response: `data/fortyguard-cache/plainview_heatmaps_2026-08-19_2026-08-25.json`
- Test crop: Upland cotton (`upland_cotton`)
- Catalog version: 1.0.0

This was an API capability and data-quality test. It was not the final 22-crop recommendation run. Cotton was selected to test a sourced crop-specific threshold; the crop-agnostic temperature result can support every eligible crop, while threshold analyses must later be repeated for the catalog's unique thresholds.

Objective

The test checked whether FortyGuard could provide usable local heat evidence for an agricultural area near Plainview, Hale County, Texas. It specifically tested:

1. Rural Texas coverage.
2. Multi-day heatmap requests using `filter_type=4`.
3. Temperature context using `analytic_type=tcm`.
4. Total hours above a crop threshold using `analytic_type=exceedance`.
5. Longest continuous exposure using `analytic_type=persistence`.
6. Hour of peak heat using `analytic_type=time_of_measure`.
7. Tile counts, numeric completeness, spatial variation, and response schemas.
8. Creation of a sanitized cached response for development and demonstrations.

Test configuration

Item	Value
Farm	Plainview demonstration farm
Representative coordinate	34.1800, -101.7600
Time zone used by CropSage	America/Chicago
Area of interest	Approximately 1 km x 1 km agricultural polygon
Requested granularity	100 metres
Date window	2026-08-19 through 2026-08-25
Crop used for threshold test	Upland cotton
Cotton heat threshold	37.8 degrees Celsius
Threshold policy	Warning with soft penalty
Threshold evidence status	Regional transfer

The 37.8 C threshold is a catalog screening threshold, not a universal biological cutoff. Crop response can vary with cultivar, growth stage, soil moisture, humidity, and farm management.

API executions

Analysis	Activity ID	Returned cells	Status
TCM temperature context	`111d22bf-84db-426d-91b6-fe872d8466d9`	68	Completed
Threshold exceedance	`b8f25479-e5e6-46b9-8744-26ddb8a8b7e9`	68	Completed
Threshold persistence	`9bb4bef3-e8b4-441e-9551-446d6190ecb4`	68	Completed
Time of peak heat	`5f4f76a7-9934-455e-9005-c169b4954178`	68	Completed

All four endpoints returned `map_data` and `stats_data`. All 68 cells in every response contained numeric values, and no null cell values were found.

Results

Temperature context

Metric	Result
Area-weighted seven-day average	29.914448 C
Lowest tile-average value	29.9143 C
Highest tile-average value	29.9146 C
Distinct tile-average values	4
Lowest recorded tile minimum	20.4795 C
Highest recorded tile minimum	20.4846 C
Distinct tile-minimum values	30
Maximum across all tiles	39.2 C

The average temperature varied by only 0.0003 C across the returned cells. The seven-day maximum was identical across all cells.

Cotton threshold exposure

Metric	Result
Cotton threshold	37.8 C
Total hours above threshold	6 hours
Longest continuous run above threshold	2 hours
Cells with the same exceedance result	68 of 68
Cells with the same persistence result	68 of 68

Six total exceedance hours with a longest continuous run of two hours means the exposure was split into shorter events. The response does not show that all six hours occurred consecutively.

Peak-time result

Metric	Result
Raw peak-hour value	15
Reported unit	hour
Cells with the same value	68 of 68
Notebook UTC interpretation	15:00 UTC
Notebook local conversion	approximately 10:00 America/Chicago
Interpretation status	Unverified

The endpoint returned an hour but did not include time-zone metadata in `stats_data`. The quickstart describes the value as UTC, but a 10:00 local peak appears unusually early for this location and season. CropSage must retain the raw value and mark its time-zone interpretation as unverified until confirmed. This field must not affect recommendation scoring yet.

Interpretation for cotton

Cotton's catalog optimal temperature band is approximately 26.7-32.2 C. The seven-day mean of 29.91 C falls within that band, so general recent temperature fit was favorable.

The 39.2 C maximum exceeded cotton's 37.8 C warning threshold. FortyGuard measured six total exceedance hours, with no continuous run longer than two hours. The correct preliminary interpretation is:

- General temperature fit: favorable.
- Heat warning: present.
- Prolonged heat exposure: limited in this window.
- Scoring action: eligible for a documented soft penalty, not automatic rejection.
- Confidence caveat: the catalog threshold is a regional-transfer value rather than a cultivar-specific local threshold.

This illustrates why CropSage must keep suitability, risk, and confidence separate. A crop can have a good overall thermal fit while still carrying a recent heat-event warning.

Spatial interpretation and map colors

The analytic heatmaps appeared as a single color because every cell returned the same exceedance, persistence, and peak-hour value. This is not missing data and does not mean that the calls failed.

The temperature averages also showed negligible spatial variation. Possible explanations include homogeneous rural conditions, provider smoothing, or an effective source resolution coarser than the displayed 100 m grid. This single test cannot distinguish among those explanations.

For this farm and time window, FortyGuard is currently supported as farm-level local and temporal heat evidence. The result does not yet justify claims about identifying meaningful within-field hot spots.

Passed checks

- API authentication and asynchronous polling worked.
- The selected rural Texas area returned data.
- Multi-day `filter_type=4` was accepted by the live API.
- All four requested analytic types completed.

- The documented analysis response field `properties.value` was present.
- TCM temperature fields were numeric and plausible in Celsius.
- Exceedance and persistence were returned in hours.
- Every request returned 68 cells with no null values.
- A sanitized 300,994-byte cache was saved without the API key.

Limitations and unresolved questions

1. The peak-hour time-zone meaning remains unverified.
2. The uniform analytic maps provide no evidence of meaningful within-field variation for this test.
3. FortyGuard cells are modeled spatial evidence, not on-farm sensors.
4. Exceedance counts hours, not degree-hours or physiological crop damage.
5. Persistence records the longest run; it does not identify every event or its exact timestamps.
6. Cotton's threshold must not be applied to other crops.
7. Crops without sourced numeric thresholds must not be assigned invented thresholds.
8. A deterministic scoring policy is still required to translate supported exposure metrics into bounded penalties.
9. The provider's short-horizon forecast remains an operational warning and cannot be used as a seasonal crop forecast.

Decision

The test supports using FortyGuard as CropSage's central recent/local heat provider. For the final engine:

- Use the TCM temperature context for all regionally eligible crops.
- Group crops by unique sourced heat threshold to avoid duplicate API calls.
- Use exceedance only where the catalog permits threshold-based interpretation.
- Use persistence as evidence and only score it if a defensible duration policy is documented.
- Preserve provider, date window, granularity, activity ID, units, cache status, and uncertainty.
- Do not use peak-hour local conversion in scoring until its time-zone semantics are confirmed.

Next test

The next notebook step will group the Texas Plains crops by their seven unique numeric thresholds and request one exceedance result per unique threshold. The results will then be mapped back to the relevant crops. All 22 catalog records remain available to the recommendation engine; regional eligibility and evidence availability determine how each record is evaluated.

Test A - All 22 Crops Over One Week

Test A status

- Status: Completed successfully
- Window: 19-25 August 2026 (7 complete days; 168 hours)
- Catalog records evaluated: 22
- Texas Plains eligible records: 21
- Regionally ineligible records retained in output: 1
- Crops with numeric heat thresholds: 10
- Unique threshold groups: 7
- FortyGuard responses validated: 16
- Returned cells per response: 68
- Null cell values across all responses: 0
- Granularity: 100 metres
- Detailed result: `data/fortyguard-cache/multi-window/2026-08-19_2026-08-25/all_crop_summary.json`

Test A reused the four completed baseline responses and made 12 new live calls. Every new response was cached immediately.

Period temperature context

Metric	Result
Seven-day mean temperature	29.9144 C
Minimum temperature	20.4795 C
Maximum temperature	39.2 C
Raw peak-hour value	15
Peak-hour time-zone interpretation	Unverified; not scored

The catalog comparison classified 12 crops as having a mean temperature within their optimal band, nine as above their optimal band, and long-grain rice as regionally ineligible for the Texas Plains. No crop was below its optimal mean band in this window.

Unique threshold results

Threshold	Crops sharing result	Exceedance	Window share	Persistence
23.9 C	Fresh-market spinach	107 h	63.69%	8 h
27.8 C	Grain oats	78 h	46.43%	8 h
29.4 C	Soybean	71 h	42.26%	8 h
30.0 C	Hard red winter wheat	69 h	41.07%	8 h
35.0 C	Grain corn, silage corn, dry-bulb onion, oilseed sunflower	45 h	26.79%	8 h
37.2 C	Grain sorghum	20 h	11.90%	3 h
37.8 C	Upland cotton	6 h	3.57%	2 h

Every one of the 68 cells returned the same exceedance and persistence value at each threshold. The threshold results are monotonic as expected: exposure falls as the threshold increases. The uniformity confirms valid farm-level temporal evidence but still does not demonstrate meaningful within-field heat differences.

Threshold activity IDs

Threshold	Exceedance activity	Persistence activity
23.9 C	fe422441-06b2-4e36-8c5b-32ff89250959	8feabc85-cda3-4303-ac4d-2788224aca9c
27.8 C	500c7514-b886-4d0a-89dc-9f3d83c5890c	4178e443-2e38-4b61-93b2-423d0d36ebd2
29.4 C	b0600d9a-928f-4d42-bee5-8f357714c2ab	ca82136c-9276-4097-8dbc-c193c5b74c75
30.0 C	7b5dc707-cb85-49bc-b49c-8932b937a133	a67dc5b5-5398-411b-b312-f76be223b2d7
35.0 C	dd35a9e1-9c07-43ad-a082-f6ea90c25ed6	e4183e26-505e-474d-8e46-664c00ad0a4a
37.2 C	39abe392-6e01-42a7-be31-409465075144	0f0350d6-f7fc-40c0-8f96-10779b45d7bc
37.8 C	b8f25479-e5e6-46b9-8744-26ddb8a8b7e9	9bb4bef3-e8b4-441e-9551-446d6190ecb4

All 22 crop outcomes

Crop	Plains eligible	Mean-temperature fit	Threshold	Exceedance	Persistence	Current heat action
Upland cotton	Yes	Within optimum	37.8 C	6 h	2 h	Soft-penalty candidate
Corn grown for grain	Yes	Within optimum	35.0 C	45 h	8 h	Soft-penalty candidate
Hard red winter wheat	Yes	Above optimum	30.0 C	69 h	8 h	Soft-penalty candidate
Grain sorghum	Yes	Within optimum	37.2 C	20 h	3 h	Soft-penalty candidate
Runner-type peanut	Yes	Within optimum	-	-	-	No numeric threshold; do not invent
Long-grain rice	No	Regionally ineligible	-	-	-	Excluded for Plains
Soybean	Yes	Above optimum	29.4 C	71 h	8 h	Soft-penalty candidate
Grain oats	Yes	Above optimum	27.8 C	78 h	8 h	Soft-penalty candidate
Oilseed sunflower	Yes	Above optimum	35.0 C	45 h	8 h	Informational warning only
Sesame	Yes	Within optimum	-	-	-	No numeric threshold; do not invent
Corn grown for silage	Yes	Within optimum	35.0 C	45 h	8 h	Soft-penalty candidate
Forage sorghum	Yes	Within optimum	-	-	-	No numeric threshold; do not invent
Sorghum-sudangrass	Yes	Within optimum	-	-	-	No numeric threshold; do not invent
Alfalfa grown for hay	Yes	Within optimum	-	-	-	No numeric threshold; do not invent
Bermudagrass grown for hay	Yes	Within optimum	-	-	-	No numeric threshold; do not invent
Annual ryegrass grown for forage	Yes	Above optimum	-	-	-	No numeric threshold; do not invent
White potato	Yes	Above optimum	-	-	-	No numeric threshold; do not invent
Sweet potato	Yes	Within optimum	-	-	-	No numeric threshold; do not invent
Seedless watermelon	Yes	Within optimum	-	-	-	No numeric threshold; do not invent
Dry-bulb onion	Yes	Above optimum	35.0 C	45 h	8 h	Informational warning only
Fresh-market cabbage	Yes	Above optimum	-	-	-	No numeric threshold; do not invent
Fresh-market spinach	Yes	Above optimum	23.9 C	107 h	8 h	Informational warning only

Test A interpretation

The one-week result differentiates crops in two complementary ways:

1. Mean-temperature fit indicates whether the overall period was inside the crop's preferred planning band.
2. Threshold exposure identifies damaging or warning-level heat events that an average can conceal.

Corn demonstrates this distinction clearly: its mean was still within the catalog optimum, but the farm spent 45 hours above 35 C. Cotton also remained within its optimum mean band but accumulated six hours above 37.8 C. These crops should retain favorable general-temperature fit while receiving separate heat-event warnings and bounded soft-penalty candidates.

Winter wheat, soybean, and oats were already above their optimal mean bands and also had substantial threshold exposure. Their heat evidence is less favorable for this late-August window. This does not by itself determine planting suitability because planting window, growth stage, soil, water, and irrigation evidence have not yet been combined.

Spinach, onion, and sunflower exceeded their numeric warning thresholds, but their catalog records permit informational use only. Their threshold results must not directly reduce the score. Their separate optimal-temperature-fit component may still identify unfavorable conditions.

The 11 eligible crops without numeric thresholds remain valid candidates. FortyGuard still supplies their period temperature context, but CropSage must not infer exceedance or persistence values for them.

Test A decision

Test A passed. The seven threshold groups behaved consistently, all response schemas validated, and the shared-threshold strategy reduced duplicate calls. The result is suitable as the one-week FortyGuard evidence layer for the later deterministic engine, subject to the documented scoring policy and time-zone limitation.

Test B - All 22 Crops Over Two Weeks

Test B status

- Status: Completed successfully
- Window: 12-25 August 2026 (14 complete days; 336 hours)
- Catalog records evaluated: 22
- Texas Plains eligible records: 21
- Regionally ineligible records retained in output: 1
- Crops with numeric heat thresholds: 10
- Unique threshold groups: 7
- New live FortyGuard responses: 16
- Returned cells per response: 68
- Null cell values across all responses: 0
- Granularity: 100 metres
- Detailed result: `data/fortyguard-cache/multi-window/2026-08-12_2026-08-25/all_crop_summary.json`

Period temperature context

Metric	Test B result	Test A result	Change
Mean temperature	29.7099 C	29.9144 C	-0.2045 C
Minimum temperature	19.77 C	20.4795 C	-0.7095 C
Maximum temperature	39.2 C	39.2 C	No change
Raw peak-hour value	15	15	No change

The time-zone meaning of peak hour 15 remains unverified and is not used for scoring. The slightly lower two-week mean did not change the catalog classification counts: 12 crops remained within their optimal mean band, nine remained above it, and long-grain rice remained regionally ineligible.

Unique threshold results

Threshold	Crops sharing result	Exceedance	Window share	Persistence
23.9 C	Fresh-market spinach	248 h	73.81%	15 h
27.8 C	Grain oats	168 h	50.00%	15 h
29.4 C	Soybean	154 h	45.83%	15 h
30.0 C	Hard red winter wheat	147 h	43.75%	15 h
35.0 C	Grain corn, silage corn, dry-bulb onion, oilseed sunflower	81 h	24.11%	15 h
37.2 C	Grain sorghum	23 h	6.85%	3 h
37.8 C	Upland cotton	6 h	1.79%	2 h

All 68 cells again returned the same analytic value at each threshold. The result remains useful as farm-level temporal evidence but does not show within-field heat differentiation.

Test A versus Test B

Because Test B contains Test A's seven days plus the earlier 12-18 August week, subtracting Test A reveals the earlier week's exposure.

Threshold	Test A share	Test B share	Added earlier-week hours	Earlier-week share
23.9 C	63.69%	73.81%	141 h	83.93%
27.8 C	46.43%	50.00%	90 h	53.57%
29.4 C	42.26%	45.83%	83 h	49.40%
30.0 C	41.07%	43.75%	78 h	46.43%
35.0 C	26.79%	24.11%	36 h	21.43%
37.2 C	11.90%	6.85%	3 h	1.79%
37.8 C	3.57%	1.79%	0 h	0.00%

The earlier week contained more sustained moderate warmth but fewer extreme-heat hours. It increased the normalized share below 30 C, but diluted the two-week share above 35 C, 37.2 C, and 37.8 C. All six cotton-threshold exceedance hours occurred in the more recent Test A week. The longest continuous run at thresholds up to 35 C increased from eight to 15 hours, while persistence at 37.2 C and 37.8 C remained unchanged.

Threshold activity IDs

Analysis	Threshold	Activity ID
TCM	-	abbec489-05db-47ec-ae25-70b257714927
Time of measure	-	93b17c9e-0009-464f-b95a-4315109e9cc9
Exceedance	23.9 C	a422202f-b304-4d4a-aa78-cd6bf6e4c692
Persistence	23.9 C	01f559fe-1a40-4ea0-a9ad-fed1e3a9de51
Exceedance	27.8 C	f0ef97db-c634-4d73-a98f-3c80e586f350
Persistence	27.8 C	a93f6764-ff41-4b27-b5ed-c165970ed327
Exceedance	29.4 C	e8a1a06f-d727-4017-b559-479924387455
Persistence	29.4 C	5007c98a-154f-41a2-9ea8-d2cc6386d96c
Exceedance	30.0 C	8b3256e0-0a91-46e7-9f75-dc0bb76583b9
Persistence	30.0 C	49239238-207f-4a69-8f6b-f3d8116e1ac2
Exceedance	35.0 C	2304e34e-8767-441e-bb22-8d5f665a586d
Persistence	35.0 C	b7663698-e7fa-4df7-abd7-9f5abda7c75b
Exceedance	37.2 C	1e8b2999-ce42-42ee-8fd0-c6ae303139fa
Persistence	37.2 C	ab0bb752-53a8-4f44-8a45-6097d51b6e82
Exceedance	37.8 C	00d60d15-3d69-43e8-8af9-ba2ad64552da
Persistence	37.8 C	98fc4df8-f5d1-40b8-b85d-d654388d7beb

All 22 crop outcomes

Crop	Plains eligible	Mean-temperature fit	Threshold	Exceedance	Persistence	Current heat action
Upland cotton	Yes	Within optimum	37.8 C	6 h	2 h	Soft-penalty candidate
Corn grown for grain	Yes	Within optimum	35.0 C	81 h	15 h	Soft-penalty candidate
Hard red winter wheat	Yes	Above optimum	30.0 C	147 h	15 h	Soft-penalty candidate
Grain sorghum	Yes	Within optimum	37.2 C	23 h	3 h	Soft-penalty candidate
Runner-type peanut	Yes	Within optimum	-	-	-	No numeric threshold; do not invent
Long-grain rice	No	Regionally ineligible	-	-	-	Excluded for Plains
Soybean	Yes	Above optimum	29.4 C	154 h	15 h	Soft-penalty candidate
Grain oats	Yes	Above optimum	27.8 C	168 h	15 h	Soft-penalty candidate
Oilseed sunflower	Yes	Above optimum	35.0 C	81 h	15 h	Informational warning only
Sesame	Yes	Within optimum	-	-	-	No numeric threshold; do not invent
Corn grown for silage	Yes	Within optimum	35.0 C	81 h	15 h	Soft-penalty candidate
Forage sorghum	Yes	Within optimum	-	-	-	No numeric threshold; do not invent
Sorghum-sudangrass	Yes	Within optimum	-	-	-	No numeric threshold; do not invent
Alfalfa grown for hay	Yes	Within optimum	-	-	-	No numeric threshold; do not invent
Bermudagrass grown for hay	Yes	Within optimum	-	-	-	No numeric threshold; do not invent
Annual ryegrass grown for forage	Yes	Above optimum	-	-	-	No numeric threshold; do not invent
White potato	Yes	Above optimum	-	-	-	No numeric threshold; do not invent
Sweet potato	Yes	Within optimum	-	-	-	No numeric threshold; do not invent
Seedless watermelon	Yes	Within optimum	-	-	-	No numeric threshold; do not invent
Dry-bulb onion	Yes	Above optimum	35.0 C	81 h	15 h	Informational warning only
Fresh-market cabbage	Yes	Above optimum	-	-	-	No numeric threshold; do not invent
Fresh-market spinach	Yes	Above optimum	23.9 C	248 h	15 h	Informational warning only

Test B interpretation

The two-week window confirms that raw exceedance hours must not be compared across different window lengths without normalization. For example, corn's exposure rose from 45 to 81 hours, but its share of the observation window fell from 26.79% to 24.11%. Cotton remained at six hours, so its exposure share fell by half when the earlier, less-extreme week was included.

The window also shows why both exceedance and persistence matter. Moderate thresholds accumulated many hours and reached 15-hour continuous runs, while the extreme 37.8 C threshold remained limited to six total hours and a two-hour continuous run.

These outputs remain evidence, not final recommendation scores. The deterministic engine must combine normalized exposure with catalog scoring permission, mean-temperature fit, planting calendar, soil, water, irrigation, and confidence.

Test B decision

Test B passed. All 16 responses validated without missing values, threshold behavior remained monotonic, shared results mapped correctly to crops, and normalized comparison exposed a meaningful difference between sustained moderate warmth and recent extreme heat. The two-week evidence is ready for later scoring-policy evaluation.

Test C - All 22 Crops Over Three Weeks

Test C status

- Status: Completed successfully
- Window: 5-25 August 2026 (21 complete days; 504 hours)
- Catalog records evaluated: 22
- Texas Plains eligible records: 21
- Crops with numeric thresholds: 10
- Unique threshold groups: 7
- Live responses: 16
- Returned cells per response: 68
- Null cell values across all responses: 0
- Detailed result: `data/fortyguard-cache/multi-window/2026-08-05_2026-08-25/all_crop_summary.json`

Test C temperature context

Metric	Result
Mean temperature	29.4247 C
Minimum temperature	19.77 C
Maximum temperature	39.2 C
Raw peak-hour value	15, time zone unverified

The mean-temperature classifications remained unchanged: 12 within optimum, nine above optimum, and long-grain rice regionally ineligible.

Test C threshold results

Threshold	Crops sharing result	Exceedance	Window share	Persistence
23.9 C	Fresh-market spinach	375 h	74.40%	22 h
27.8 C	Grain oats	258 h	51.19%	22 h
29.4 C	Soybean	232 h	46.03%	22 h
30.0 C	Hard red winter wheat	225 h	44.64%	22 h
35.0 C	Grain corn, silage corn, dry-bulb onion, oilseed sunflower	107 h	21.23%	22 h
37.2 C	Grain sorghum	23 h	4.56%	3 h
37.8 C	Upland cotton	6 h	1.19%	2 h

The added 5-11 August week increased moderate-threshold exposure but added no hours above 37.2 C or 37.8 C. Consequently, the normalized extreme-heat share fell again even though the observation window grew.

Test C activity IDs

Analysis	Threshold	Activity ID
TCM	-	69744abd-db6f-4c4c-aa82-df20bbb2ba79
Time of measure	-	2c4eda74-fe34-4dde-a8c9-937e30d0a1d3
Exceedance	23.9 C	405c0a11-30a6-44fc-8e92-9854682ac75d
Persistence	23.9 C	70bfada6-d5b3-4f64-8ad0-b73aa148611d
Exceedance	27.8 C	9517612d-741a-4b03-8f76-bf5d53f6f8ec
Persistence	27.8 C	bf46b3f8-7d28-4fcd-a546-dcb0bc25ade6
Exceedance	29.4 C	1758e1e7-692c-4800-bc71-2dd9eb33eacb
Persistence	29.4 C	5b2fca90-b144-4bc7-9239-704511908c32
Exceedance	30.0 C	715a0dbe-9724-416d-9365-286188816206
Persistence	30.0 C	e983ccf4-ec85-4260-a69e-3362fd932d9f
Exceedance	35.0 C	e76b1f2e-88cd-4d38-8b1e-b2046c89f7bd
Persistence	35.0 C	9e05648e-86b8-41eb-9d18-094b2c5e653f
Exceedance	37.2 C	bbedcb29-33f5-4362-aa97-edfbb5de3699
Persistence	37.2 C	858b363b-0211-4b28-8134-6a4553d563ae
Exceedance	37.8 C	7c5394c7-1a2d-4d40-9d02-8d6b607352bd
Persistence	37.8 C	91502d89-0f18-40eb-8863-5b7707a3ec18

Test C decision

Test C passed. It reinforces that late-period extreme exposure was concentrated in the most recent seven days, while moderate heat was sustained throughout the broader August window.

Test D - All 22 Crops Over 30 Days

Test D status

- Status: Completed successfully
- Window: 27 July-25 August 2026 (30 complete days; 720 hours)
- Catalog records evaluated: 22
- Texas Plains eligible records: 21
- Crops with numeric thresholds: 10
- Unique threshold groups: 7
- Live responses: 16
- Returned cells per response: 68
- Null cell values across all responses: 0
- API window acceptance: 30-day `filter_type=4` succeeded
- Detailed result: `data/fortyguard-cache/multi-window/2026-07-27_2026-08-25/all_crop_summary.json`

Test D temperature context

Metric	Result
Mean temperature	29.4330 C
Minimum temperature	19.4 C
Maximum temperature	39.2 C
Raw peak-hour value	15, time zone unverified

The all-crop mean-temperature classifications again remained unchanged: 12 within optimum, nine above optimum, and long-grain rice regionally ineligible.

Test D threshold results

Threshold	Crops sharing result	Exceedance	Window share	Persistence
23.9 C	Fresh-market spinach	549 h	76.25%	31 h
27.8 C	Grain oats	379 h	52.64%	31 h
29.4 C	Soybean	340 h	47.22%	31 h
30.0 C	Hard red winter wheat	328 h	45.56%	31 h
35.0 C	Grain corn, silage corn, dry-bulb onion, oilseed sunflower	153 h	21.25%	24 h
37.2 C	Grain sorghum	27 h	3.75%	3 h
37.8 C	Upland cotton	6 h	0.83%	2 h

The additional nine days before Test C added 174 hours above 23.9 C, 121 above 27.8 C, 108 above 29.4 C, 103 above 30 C, 46 above 35 C, four above 37.2 C, and none above 37.8 C. The added period therefore contained sustained moderate-to-high heat but no cotton-threshold event.

Test D activity IDs

Analysis	Threshold	Activity ID
TCM	-	05f0a138-6ff3-4c57-825a-499517eb3e70
Time of measure	-	8825c549-ae32-4fe4-9ff5-4b467e919fb1
Exceedance	23.9 C	7cb7decb-26e3-4bc7-8972-070c3d4d19a1
Persistence	23.9 C	3c050d8e-adef-40cd-820f-4c747e8017ab
Exceedance	27.8 C	a224772c-505b-4064-9a2b-db4d90243381
Persistence	27.8 C	809cbdf5-d05a-4b87-b505-08167322b95c
Exceedance	29.4 C	0240496c-56ba-416c-b718-74fb4fe3b92d
Persistence	29.4 C	944c7557-f838-4b10-911f-10320c8eed34
Exceedance	30.0 C	8a5f052b-0b6f-49d3-b983-644b9d55149b
Persistence	30.0 C	f95b0cff-a52a-41dc-88fc-2afca7300bf6
Exceedance	35.0 C	1d66af80-ed02-4758-9038-e83337c5bbe3
Persistence	35.0 C	bb001205-7c26-4993-b7e7-aa062c016a72
Exceedance	37.2 C	7b55b8df-bc67-4c82-a8f4-9178729b1ba9
Persistence	37.2 C	5c3cc7d2-3bcc-4b2b-99fb-a8509e9f6338
Exceedance	37.8 C	bfdc8689-57df-4b06-82cd-54113a24245b
Persistence	37.8 C	2e6b17e7-60d4-4f56-824c-4c7fb39fcad4

Combined all-crop outcomes for Tests C and D

Crop	Mean-temperature fit	Threshold	Test C exposure / persistence	Test D exposure / persistence	Heat action
Upland cotton	Within optimum	37.8 C	6 h / 2 h	6 h / 2 h	Soft-penalty candidate
Corn grown for grain	Within optimum	35.0 C	107 h / 22 h	153 h / 24 h	Soft-penalty candidate
Hard red winter wheat	Above optimum	30.0 C	225 h / 22 h	328 h / 31 h	Soft-penalty candidate
Grain sorghum	Within optimum	37.2 C	23 h / 3 h	27 h / 3 h	Soft-penalty candidate
Runner-type peanut	Within optimum	-	-	-	No numeric threshold; do not invent
Long-grain rice	Regionally ineligible	-	-	-	Excluded for Plains
Soybean	Above optimum	29.4 C	232 h / 22 h	340 h / 31 h	Soft-penalty candidate
Grain oats	Above optimum	27.8 C	258 h / 22 h	379 h / 31 h	Soft-penalty candidate
Oilseed sunflower	Above optimum	35.0 C	107 h / 22 h	153 h / 24 h	Informational warning only
Sesame	Within optimum	-	-	-	No numeric threshold; do not invent
Corn grown for silage	Within optimum	35.0 C	107 h / 22 h	153 h / 24 h	Soft-penalty candidate
Forage sorghum	Within optimum	-	-	-	No numeric threshold; do not invent
Sorghum-sudangrass	Within optimum	-	-	-	No numeric threshold; do not invent
Alfalfa grown for hay	Within optimum	-	-	-	No numeric threshold; do not invent
Bermudagrass grown for hay	Within optimum	-	-	-	No numeric threshold; do not invent
Annual ryegrass grown for forage	Above optimum	-	-	-	No numeric threshold; do not invent
White potato	Above optimum	-	-	-	No numeric threshold; do not invent
Sweet potato	Within optimum	-	-	-	No numeric threshold; do not invent
Seedless watermelon	Within optimum	-	-	-	No numeric threshold; do not invent
Dry-bulb onion	Above optimum	35.0 C	107 h / 22 h	153 h / 24 h	Informational warning only
Fresh-market cabbage	Above optimum	-	-	-	No numeric threshold; do not invent
Fresh-market spinach	Above optimum	23.9 C	375 h / 22 h	549 h / 31 h	Informational warning only

Test D decision

Test D passed. The API accepted the full 30-day range, all response schemas validated, and the result supplies the longest supported recent-history comparison required by the current test plan.

Cross-Window A-D Summary

Overall validation

Validation item	Result
Test windows	4
Catalog evaluations	88 crop-window rows
Validated FortyGuard responses	64
Cells per response	68
Numeric cell values checked	4,352
Null values	0
Unique threshold groups per window	7
Failed requests	0

Temperature context across windows

Test	Days	Mean	Minimum	Maximum	Raw peak hour
A	7	29.9144 C	20.4795 C	39.2 C	15
B	14	29.7099 C	19.77 C	39.2 C	15
C	21	29.4247 C	19.77 C	39.2 C	15
D	30	29.4330 C	19.4 C	39.2 C	15

The broader-window means remained stable within about 0.49 C, while the same 39.2 C extreme was retained in every nested window.

Normalized threshold exposure across windows

Threshold	Test A 7d	Test B 14d	Test C 21d	Test D 30d
23.9 C	63.69%	73.81%	74.40%	76.25%
27.8 C	46.43%	50.00%	51.19%	52.64%
29.4 C	42.26%	45.83%	46.03%	47.22%
30.0 C	41.07%	43.75%	44.64%	45.56%
35.0 C	26.79%	24.11%	21.23%	21.25%
37.2 C	11.90%	6.85%	4.56%	3.75%
37.8 C	3.57%	1.79%	1.19%	0.83%

The nested windows reveal two different patterns. Moderate warmth below 30 C was common throughout the month, so its normalized share increased as earlier dates were added. Extreme heat above 37.2 C was concentrated near the end of the window, so its normalized share declined sharply as the observation window expanded.

Persistence across windows

Threshold	Test A 7d	Test B 14d	Test C 21d	Test D 30d
23.9 C	8 h	15 h	22 h	31 h
27.8 C	8 h	15 h	22 h	31 h
29.4 C	8 h	15 h	22 h	31 h
30.0 C	8 h	15 h	22 h	31 h
35.0 C	8 h	15 h	22 h	24 h
37.2 C	3 h	3 h	3 h	3 h
37.8 C	2 h	2 h	2 h	2 h

The extreme-event duration did not grow when earlier dates were added. This supports treating the 37.2 C and 37.8 C results as recent, short extreme events rather than month-long persistent exposure.

Final multi-window decision

All four tests passed technically and produced internally consistent threshold behavior. The final scoring engine should use both raw exposure and normalized exposure share, because raw hours cannot be compared directly across window lengths. Persistence should remain a separate risk signal, and no penalty amount should be assigned until the deterministic scoring policy is documented.

The catalog outcome categories remained stable across all four windows:

- 12 crops within their optimal mean-temperature band.
- Nine crops above their optimal mean-temperature band.
- Seven soft-penalty candidates with supported numeric thresholds.
- Three informational threshold warnings.
- Eleven eligible crops without numeric thresholds.
- One crop regionally excluded for the Texas Plains.

All analytic heatmaps remained spatially uniform across the 68 cells. FortyGuard is therefore supported here as local farm-level temporal heat evidence, but these tests do not support claims of detecting within-field hot spots.